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```sql
--1 Tampilkan nama pelanggan dan total belanja dari semua pesanan mereka
select c.customer_name, sum(oi.subtotal) as total_belanja
from customers c
join orders o on c.customer_id = o.customer_id
join order_items oi on o.order_id = oi.order_id
group by customer_name;
```

	customer_name	total_belanja
1	Paula Shaw	11,506.69
2	Mark Torres	2,180.54
3	Judith Harris	7,050.5703
4	Sharon Ho	4,808.82
5	Victoria Weeks	2,981.75
6	Harry Griffin	5,481.23
7	Alison Bradshaw	7,761.9004
8	Sharon Brown	3,644.9
9	Kellie Ellison	3,627.6802
10	Ralph Rodriguez	1,261.8401
11	Robert Krueger	8,240.66
12	Ashley Gallagher	7,754.1694
13	Michael Castillo	3,612.49
14	Andrea Jackson	11,904.08
15	Amanda Zimmerman	2,406.33
16	Adam Thompson	163.16
17	Scott Moore	646.05
18	Scott Hernandez	14,183.13
19	James Brandt	9,874.4
20	Christine Long	3,467.22
21	Ashley Chen	6,267.37
22	Lori Berry	2,768.49
23	Daniel Miller	661.74
24	Joshua Hall	7,348.08
25	Calvin Young	13,625.98
26	Becky Hamilton	5,513.16

```sql
--2 Tampilkan nama produk yang dijual oleh setiap pemasok.
select p.product_name, s.supplier_name
from suppliers s
join product_suppliers ps on s.supplier_id = ps.supplier_id
join products p on ps.product_id = p.product_id;
```

	product_name	supplier_name
1	Debate	Bailey, Floyd and Bradley
2	Similar	Barnes, King and Goodman
3	Similar	Bailey, Floyd and Bradley
4	Design	Barnes, King and Goodman
5	Field	Brown, Daniel and Johnson
6	Begin	Larsen, Howard and Norton
7	Glass	Bailey, Floyd and Bradley
8	Debate	Barnes, King and Goodman
9	Finally	Barnes, King and Goodman
10	Those	Baker, White and Smith
11	Anyone	Baker, White and Smith
12	Affect	Huynh Inc
13	Nation	Brown, Daniel and Johnson
14	Debate	Brown, Daniel and Johnson
15	Truth	Owens-Singh
16	Truth	Bailey, Floyd and Bradley
17	Especially	Brown, Daniel and Johnson
18	Begin	Huynh Inc
19	Great	Owens-Singh
20	Drug	Brown, Daniel and Johnson
21	Drug	Baker, White and Smith
22	Responsibility	Martin, Estrada and Riley
23	Anyone	Bailey, Floyd and Bradley
24	Out	Barnes, King and Goodman
25	Happen	Burton-Carson
26	Field	Huynh Inc
27	Street	Burton-Carson
28	Listen	Larsen, Howard and Norton
29	Different	Fox PLC
30	Tough	Brown, Daniel and Johnson
31	Financial	Baker, White and Smith
32	See	Burton-Carson
33	Where	Barnes, King and Goodman
34	Out	Larsen, Howard and Norton
35	Anyone	Fox PLC
36	Through	Fox PLC
37	Out	Martin, Estrada and Riley
38	Professional	Burton-Carson
39	Despite	Larsen, Howard and Norton
40	Office	Martin, Estrada and Riley
41	Street	Bailey, Floyd and Bradley
42	Many	Brown, Daniel and Johnson
43	Glass	Owens-Singh
44	Body	Martin, Estrada and Riley
45	Somebody	Brown, Daniel and Johnson
46	Great	Fox PLC
47	Responsibility	Owens-Singh
48	Professional	Burton-Carson
49	Despite	Burton-Carson
50	When	Brown, Daniel and Johnson

```sql
--3 Tampilkan produk yang tidak pernah dipesan
select p.product_name
from products p
left join order_items oi on p.product_id = oi.product_id
where oi.product_id is null;
```

	product_name
1	Final
2	Listen
3	Design
4	Financial

```sql
--4 Berapa rata-rata harga produk untuk setiap kategori?
select p.category, avg(p.price) as rata_rata
from products p
group by p.category;
```

	category	rata_rata
1	Home Appliances	561.5463652177
2	Electronics	508.1300079346
3	Toys	506.0699999673
4	Clothing	421.5336403413
5	Books	537.8009109497

```sql
--5 Tampilkan pelanggan yang total pembelian mereka melebihi rata-rata total pembelian dari semua pelanggan
select c.customer_name, sum(oi.subtotal) as tot_lebih_ratarata
from customers c
join orders o on c.customer_id = o.customer_id
join order_items oi on o.order_id = oi.order_id
group by c.customer_name
having
    sum(oi.subtotal) >
    (
    select avg(tot_lebih_ratarata)
    from
        (
        select sum(oi.subtotal) as tot_lebih_ratarata
        from customers c
        join orders o on c.customer_id = o.customer_id
        join order_items oi on o.order_id = oi.order_id
        group by c.customer_id
        )
    );
```

	customer_name	tot_lebih_ratarata
1	Paula Shaw	11,506.69
2	Judith Harris	7,050.5703
3	Alison Bradshaw	7,761.9004
4	Robert Krueger	8,240.66
5	Ashley Gallagher	7,754.1694
6	Andrea Jackson	11,904.08
7	Scott Hernandez	14,183.13
8	James Brandt	9,874.4
9	Ashley Chen	6,267.37
10	Joshua Hall	7,348.08
11	Calvin Young	13,625.98

```sql
--6 Tampilkan kategori produk yang total penjualannya di atas rata-rata total penjualan semua kategori.
with tab_totalcat as
    (
    select p.category, sum(oi.subtotal) as total_penjualan
    from products p
    join order_items oi on oi.product_id = p.product_id
    group by p.category
    )
select category, total_penjualan
from tab_totalcat
where total_penjualan >
    (
    select avg(total_penjualan)
    from tab_totalcat
    );
```

	category	total_penjualan
1	Home Appliances	52,041.535
2	Clothing	35,902.09
3	Books	31,841.201

```sql
--7 Tampilkan produk yang disuplai oleh lebih dari satu pemasok dan hitung jumlah pemasok untuk setiap produk.
select p.product_name , count(ps.supplier_id) as jumlah_pemasok
from products p
join product_suppliers ps on p.product_id = ps.product_id
group by p.product_name
having count(ps.supplier_id) > 1;
```

	product_name	jumlah_pemasok
1	Truth	2
2	Great	2
3	Out	3
4	Debate	3
5	Glass	2
6	Street	2
7	Begin	2
8	Professional	2
9	Anyone	3
10	Similar	2
11	Field	2
12	Drug	2
13	Despite	2
14	Responsibility	2

```sql
--8 Tampilkan nama pelanggan, nama produk, dan tanggal pesanan untuk semua
--pesanan di mana pelanggan memesan lebih dari 2 unit produk tertentu.
select c.customer_name, p.product_name, o.order_date
from customers c
join orders o on c.customer_id = o.customer_id
join order_items oi on o.order_id = oi.order_id
join products p on oi.product_id = p.product_id
where quantity > 2;
```

	customer_name	product_name	order_date
1	Andrea Jackson	Begin	27/02/2023
2	Robert Krueger	Out	14/08/2024
3	Becky Hamilton	Field	11/03/2024
4	Christine Long	Coach	26/05/2024
5	Calvin Young	Head	10/02/2023
6	Calvin Young	Somebody	10/02/2023
7	Scott Hernandez	Professional	21/10/2023
8	James Brandt	Those	21/08/2024
9	James Brandt	Effect	21/08/2024
10	Ashley Gallagher	Effect	05/08/2024
11	Ashley Gallagher	Field	27/03/2023
12	Ashley Chen	Begin	08/10/2024
13	Ashley Chen	Himself	08/10/2024
14	Scott Hernandez	Begin	16/04/2023
15	Ralph Rodriguez	When	15/05/2024
16	Calvin Young	Glass	23/05/2024
17	Paula Shaw	Nation	14/11/2023
18	Paula Shaw	Body	14/11/2023
19	Paula Shaw	Glass	14/11/2023
20	Paula Shaw	Office	14/11/2023
21	Michael Castillo	Coach	03/08/2024
22	Amanda Zimmerman	Body	22/01/2023
23	Amanda Zimmerman	Street	22/01/2023
24	Scott Moore	Where	06/11/2024
25	Scott Moore	Truth	06/11/2024
26	Andrea Jackson	Similar	12/11/2022
27	Daniel Miller	Truth	19/08/2023
28	Alison Bradshaw	Somebody	10/01/2023
29	Sharon Brown	Drug	10/09/2024
30	Sharon Brown	Drug	10/09/2024
31	Harry Griffin	Different	21/04/2024
32	Judith Harris	Begin	12/09/2023
33	Judith Harris	Himself	12/09/2023
34	Scott Hernandez	Thought	04/02/2023
35	Scott Hernandez	Especially	04/02/2023
36	Lori Berry	Close	16/02/2023
37	Sharon Ho	Different	27/04/2023
38	Sharon Ho	Despite	27/04/2023
39	Ralph Rodriguez	Thought	18/05/2023
40	Robert Krueger	Anyone	20/12/2022
41	Robert Krueger	Nation	20/12/2022
42	Robert Krueger	Through	20/12/2022
43	Robert Krueger	Expert	20/12/2022
44	Calvin Young	Where	27/09/2023
45	Calvin Young	Where	27/09/2023
46	Andrea Jackson	Affect	08/10/2024
47	Andrea Jackson	Professional	08/10/2024
48	Victoria Weeks	Tough	05/11/2023
49	Victoria Weeks	Nation	05/11/2023
50	Alison Bradshaw	Out	19/11/2022
51	Alison Bradshaw	Nation	19/11/2022
52	Alison Bradshaw	Happen	19/11/2022
53	Mark Torres	Debate	02/12/2022
54	Mark Torres	Similar	02/12/2022
55	Mark Torres	Reach	02/12/2022
56	Robert Krueger	Peace	01/07/2024
57	Robert Krueger	Great	01/07/2024
58	Joshua Hall	Begin	25/01/2023
59	Joshua Hall	Office	25/01/2023

```sql
--9 Tampilkan nama pemasok, nama produk yang mereka suplai, serta jumlah unit
--produk tersebut yang telah dipesan oleh setiap pelanggan.
select s.supplier_name, p.product_name, sum(oi.quantity) as jumlah_unit
from suppliers s
join product_suppliers ps on s.supplier_id = ps.supplier_id
join products p on ps.product_id = p.product_id
join order_items oi on p.product_id = oi.product_id
group by s.supplier_name, p.product_name
order by s.supplier_name;
```

	supplier_name	product_name	jumlah_unit
1	Bailey, Floyd and Bradley	Anyone	3
2	Bailey, Floyd and Bradley	Debate	4
3	Bailey, Floyd and Bradley	Glass	9
4	Bailey, Floyd and Bradley	Similar	9
5	Bailey, Floyd and Bradley	Street	7
6	Bailey, Floyd and Bradley	Truth	6
7	Baker, White and Smith	Anyone	3
8	Baker, White and Smith	Drug	8
9	Baker, White and Smith	Those	5
10	Barnes, King and Goodman	Debate	4
11	Barnes, King and Goodman	Finally	2
12	Barnes, King and Goodman	Out	7
13	Barnes, King and Goodman	Similar	9
14	Barnes, King and Goodman	Where	17
15	Brown, Daniel and Johnson	Debate	4
16	Brown, Daniel and Johnson	Drug	8
17	Brown, Daniel and Johnson	Especially	5
18	Brown, Daniel and Johnson	Field	9
19	Brown, Daniel and Johnson	Many	4
20	Brown, Daniel and Johnson	Nation	17
21	Brown, Daniel and Johnson	Somebody	8
22	Brown, Daniel and Johnson	Tough	4
23	Brown, Daniel and Johnson	When	4
24	Burton-Carson	Despite	3
25	Burton-Carson	Happen	4
26	Burton-Carson	Professional	20
27	Burton-Carson	See	2
28	Burton-Carson	Street	7
29	Fox PLC	Anyone	3
30	Fox PLC	Different	10
31	Fox PLC	Great	5
32	Fox PLC	Through	5
33	Huynh Inc	Affect	6
34	Huynh Inc	Begin	11
35	Huynh Inc	Field	9
36	Larsen, Howard and Norton	Begin	11
37	Larsen, Howard and Norton	Despite	3
38	Larsen, Howard and Norton	Out	7
39	Martin, Estrada and Riley	Body	8
40	Martin, Estrada and Riley	Office	13
41	Martin, Estrada and Riley	Out	7
42	Martin, Estrada and Riley	Responsibility	1
43	Owens-Singh	Glass	9
44	Owens-Singh	Great	5
45	Owens-Singh	Responsibility	1
46	Owens-Singh	Truth	6

```sql
--10 Tampilkan nama pelanggan, nama produk, dan jumlah total yang dipesan untuk
--setiap produk oleh pelanggan yang memiliki total pembelian di atas rata-rata dari semua pelanggan.
with tabell as
    (
    select c.customer_id, c.customer_name, sum(oi.subtotal) as total_pembelian
    from customers c
    join orders o on c.customer_id = o.customer_id
    join order_items oi on o.order_id = oi.order_id
    join products p on oi.product_id = p.product_id
    group by c.customer_id, c.customer_name
    ), diatas_rata2 as
    (
    select customer_id, customer_name, total_pembelian
    from tabell
    where total_pembelian >
            (select avg(total_pembelian) as rerata
            from tabell)
    )
select dt.customer_name, p.product_name, sum(oi.subtotal) as total_per_produk
from diatas_rata2 dt
join orders o on dt.customer_id = o.customer_id
join order_items oi on o.order_id = oi.order_id
join products p on oi.product_id = p.product_id
group by dt.customer_name, p.product_name
order by dt.customer_name, p.product_name
```

	customer_name	product_name	total_per_produk
1	Alison Bradshaw	Affect	1,413.87
2	Alison Bradshaw	Happen	882.01
3	Alison Bradshaw	Nation	317.99
4	Alison Bradshaw	Out	2,237.64
5	Alison Bradshaw	Somebody	1,941.3
6	Alison Bradshaw	Street	969.09
7	Andrea Jackson	Affect	1,908.24
8	Andrea Jackson	Begin	1,946.84
9	Andrea Jackson	Office	3,180.4
10	Andrea Jackson	Professional	454.21
11	Andrea Jackson	Response	1,210.28
12	Andrea Jackson	See	1,232.78
13	Andrea Jackson	Similar	1,971.33
14	Ashley Chen	Affect	74.05
15	Ashley Chen	Begin	1,971.33
16	Ashley Chen	Himself	3,528.04
17	Ashley Chen	Peace	693.95
18	Ashley Gallagher	Effect	1,834.7
19	Ashley Gallagher	Expert	4,614.15
20	Ashley Gallagher	Field	1,305.32
21	Calvin Young	Begin	3,469.75
22	Calvin Young	Glass	1,738.12
23	Calvin Young	Head	958.25
24	Calvin Young	Somebody	376.01
25	Calvin Young	Station	2,018.2
26	Calvin Young	Where	5,065.65
27	James Brandt	Effect	4,113.75
28	James Brandt	Finally	1,210.28
29	James Brandt	Great	2,857.56
30	James Brandt	Office	917.35
31	James Brandt	Those	775.46
32	Joshua Hall	Begin	922.7
33	Joshua Hall	Office	3,968.08
34	Joshua Hall	Ten	2,457.3
35	Judith Harris	Begin	3,654.6
36	Judith Harris	Himself	784.5
37	Judith Harris	Station	154.17
38	Judith Harris	Ten	2,457.3
39	Paula Shaw	Body	636.22
40	Paula Shaw	Glass	942.58
41	Paula Shaw	Many	2,468.25
42	Paula Shaw	Nation	1,504.04
43	Paula Shaw	Office	2,073
44	Paula Shaw	Station	3,882.6
45	Robert Krueger	Anyone	4,762.6
46	Robert Krueger	Expert	388.76
47	Robert Krueger	Great	686.37
48	Robert Krueger	Maybe	1,588.54
49	Robert Krueger	Nation	261.5
50	Robert Krueger	Out	388.76
51	Robert Krueger	Peace	23.94
52	Robert Krueger	Thought	95.76
53	Robert Krueger	Through	44.43
54	Scott Hernandez	Begin	3,181.1
55	Scott Hernandez	Close	4,568.25
56	Scott Hernandez	Especially	443.76
57	Scott Hernandez	Professional	3,729.4
58	Scott Hernandez	Street	411.45
59	Scott Hernandez	Thought	1,849.17